Value investing: buy-and-hold results

S&P MidCap 400, data as of 2026-10-31

Abstract

Each January and July since 2016, a rule-based value score picks 10 S&P MidCap 400 companies, and 100 USD is invested in them, weighted by score and never sold. Investors who started in 2016, 2020, 2023 and 2025 (an earlier start buys from 2016) are followed to 30 October 2026, each against the same money in the iShares Core S&P Mid-Cap fund (IJH), dividends reinvested, net of its 0.05% yearly fee, and in Berkshire Hathaway. The reference is the investor who started in 2020: the data hold only today's index members, and by the index's list of changes, which starts in January 2012, 293 companies have left the index since then against 468 since 2016, so fewer failed companies are missing. That investor's 1,400 USD is worth 2,318 USD, against 2,073 USD in the benchmark and 2,415 USD in Berkshire Hathaway; a second wallet that also sells holdings whose score has fallen is worth 2,361 USD. The report closes with the next picks: UFP Industries (UFPI), Harley-Davidson (HOG), Ingredion (INGR), Simpson Manufacturing (SSD), NOV Inc. (NOV), Fluor (FLR), Commercial Metals (CMC), Avnet (AVT), Visteon (VC) and InterDigital (IDCC). The result rests on few decisions and few companies, with 14% of the value in one company; it is a research result, not evidence of a lasting edge.

1 Method

Company fundamentals are the values first reported in each annual report (10-K) to the SEC, used from the day they were filed. When a company reports an item under more than one label, the label it used most recently by the date is taken, so later filings never decide it (Appendix D). Prices and dividends are from Yahoo Finance; returns include reinvested dividends and are in US dollars. On each date, only companies already in the S&P MidCap 400 are considered, outside the Financials and Real Estate sectors.

Ten rules from Graham and Buffett grade every company: G6 and G7 (price against earnings and book value), G2a and E1 (current ratio), G2b and E2 (debt), NN (net-net), G5 (earnings growth), B1 (return on equity) and E4 (current dividend). Six more (G3, G4, E3, E5, E6 and NI1) are switched off and can be switched on. The ten are eight concepts, because G2a and E1 (current ratio) and G2b and E2 (debt cover) measure the same thing and so count twice. For each rule, the companies are ranked by the rule's value and placed in equal-sized groups, from the most points for the best group to none for the worst; each rule's points are multiplied by its weight, one less the share of companies that pass the rule's threshold, so rules few companies pass count more. The score is the sum. On the first trading day of January and of July, the 10 highest-scoring companies are bought for 100 USD, each in proportion to its score.

Three limits apply to every number in this report. Companies that left the index are missing from the data, which flatters the results. The score's design and every setting of this report (the rules and their weighting, the number of picks, the decision months, the sell threshold, the start years) were chosen while looking at the whole period from 2009, so no period here is out of sample: the results describe how these rules would have done, not evidence that they will do so. There are no trading costs or taxes.

Neither Benjamin Graham nor Berkshire Hathaway or Warren Buffett has endorsed, reviewed or approved this score or this report. The rules are our reading of Graham's books; the quotations from Berkshire's published letters to shareholders are cited by year and show where a question was taken, not what any of them would say about these results.

2 Picks per half-year

Table 1: Purchases per half-year and their value today (USD).

Bought	Picks and weights	Wallet	S&P MidCap 400	Berkshire
2016-01	PPC 12%, LECO 11%, CMC 10%, GNTX 10%, AVT 10%, TKR 10%, GAP 9%, RRX 9%, VMI 9%, LSTR 9%	349	313	415
2016-07	GNTX 13%, BC 11%, ITT 11%, AVT 10%, SNX 10%, GAP 10%, CMC 10%, CXT 9%, GHC 8%, PPC 8%	324	285	377
2017-01	GNTX 13%, BC 11%, ITT 11%, CMC 11%, AVT 10%, GAP 9%, SNX 9%, PPC 9%, MMS 9%, GHC 9%	254	254	331
2017-07	GNTX 13%, CMC 11%, AVT 11%, ITT 10%, SNX 10%, GAP 10%, ENS 9%, UTHR 9%, SYNA 9%, LEA 9%	288	240	316
2018-01	GNTX 13%, AVT 11%, SYNA 11%, ENS 10%, MMS 10%, SNX 9%, GAP 9%, UTHR 9%, ARW 9%, ITT 9%	249	218	275
2018-07	GNTX 12%, AVT 11%, LPX 11%, MMS 10%, SYNA 10%, MKSI 10%, ARW 10%, GAP 9%, UTHR 9%, CRUS 9%	237	212	289
2019-01	GNTX 12%, LPX 11%, MKSI 11%, AVT 10%, MMS 10%, OSK 9%, VC 9%, GAP 9%, ARW 9%, INGR 9%	217	248	267
2019-07	LPX 12%, MKSI 12%, GNTX 11%, AVT 10%, GAP 10%, FORM 9%, INGR 9%, RS 9%, ARW 9%, OSK 9%	271	209	253
2020-01	LPX 12%, MKSI 11%, AYI 10%, GAP 10%, AVT 10%, ARW 9%, LSTR 9%, INGR 9%, RS 9%, MSM 9%	225	196	238
2020-07	AYI 12%, AVT 11%, RS 10%, LSTR 10%, AVNT 10%, TKR 10%, MSM 9%, GGG 9%, MTZ 9%, INGR 9%	287	227	305
2021-01	LSTR 11%, RS 11%, AYI 10%, CMC 10%, MMS 10%, JAZZ 10%, INGR 10%, MSM 10%, LEA 9%, GGG 9%	190	176	237
2021-07	LPX 11%, LOPE 11%, FORM 11%, RS 10%, LSTR 10%, AYI 10%, GGG 10%, CMC 10%, MMS 9%, TKR 9%	167	146	194
2022-01	CMC 12%, OLLI 11%, LOPE 11%, FORM 10%, TKR 10%, MKSI 10%, LPX 10%, AVT 9%, GGG 9%, JAZZ 9%	161	138	180
2022-07	LOPE 12%, RS 11%, CMC 11%, MKSI 10%, RH 10%, LPX 10%, TKR 9%, ARW 9%, GGG 9%, LSTR 9%	171	170	195
2023-01	AVT 12%, RS 11%, CMC 11%, MKSI 11%, ARW 10%, RH 9%, IPGP 9%, LPX 9%, GGG 9%, OLLI 9%	164	160	175
2023-07	UFPI 12%, RS 11%, AVT 10%, CMC 10%, ARW 10%, LPX 10%, AMKR 9%, SSD 9%, YETI 9%, RH 9%	127	146	159
2024-01	CMC 12%, UFPI 12%, RS 11%, AVT 10%, ARW 10%, LPX 10%, MSM 9%, LSTR 9%, AMKR 9%, ASH 9%	129	137	150
2024-07	RS 11%, UFPI 11%, CMC 11%, AVT 11%, ARW 10%, TEX 10%, ROIV 10%, MSM 9%, YETI 9%, LSTR 9%	154	130	134
2025-01	AVT 11%, CMC 11%, RS 11%, UFPI 11%, MLI 10%, ARW 10%, TEX 10%, AMKR 9%, ROIV 9%, FN 9%	161	121	120
2025-07	AVT 11%, CMC 11%, MLI 11%, UFPI 11%, ROIV 10%, RS 10%, ARW 9%, NOV 9%, YETI 9%, VC 9%	151	119	111
2026-01	UFPI 12%, AVT 11%, MLI 10%, RS 10%, ARW 10%, VC 10%, NOV 10%, LSTR 10%, INGR 9%, SSD 9%	121	110	109
2026-07	UFPI 11%, AVT 11%, MLI 10%, HOG 10%, INGR 10%, SSD 10%, NOV 10%, ANF 9%, IDCC 9%, FLR 9%	110	96	109

Table 1 lists the companies bought each January and July, with their weights, and what each 100 USD is worth on 30 October 2026, next to the same money in the S&P MidCap 400 and in Berkshire Hathaway. Figure 1 shows the same values. Most of the money went to Industrials, Information Technology and Consumer Discretionary; the best single purchase is ITT in July 2016, now worth 7 times its cost. The purchases of January 2017, January 2019, July 2023 and January 2024 are worth less today than the same money in the S&P MidCap 400. Appendix A lists every purchase of the wallet with sells of the investor who started in 2020, with its price at purchase and today,

what later sales took from it, what it keeps today and its total return with dividends.

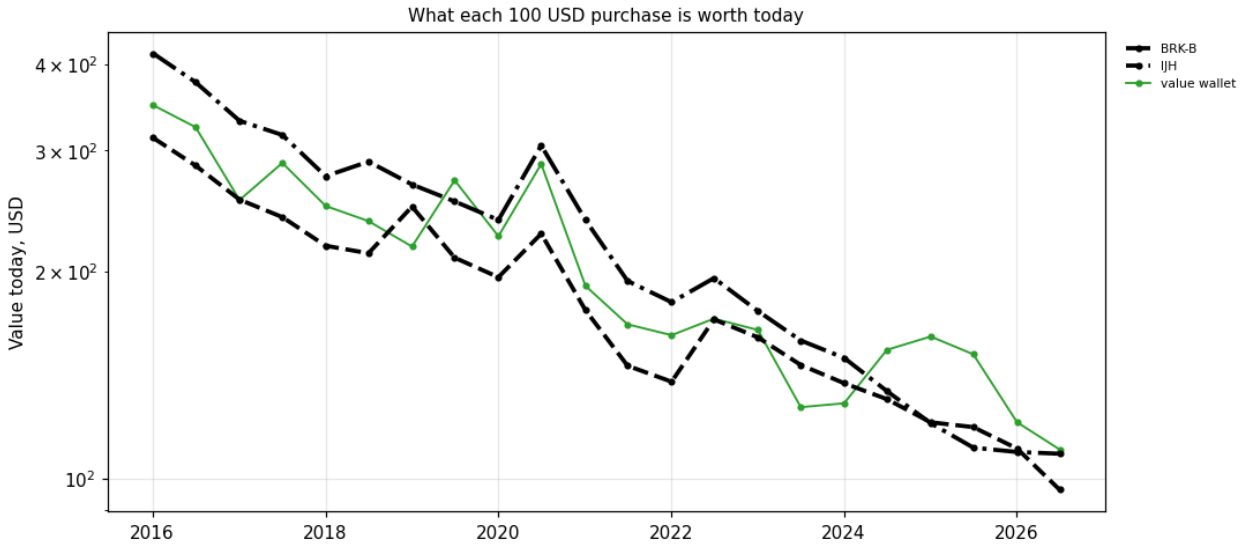


Figure 1: What each purchase is worth today, by date of purchase, on a logarithmic scale.

3 Value over time

Table 2: Money put in and value today per start year (USD), and the money-weighted return a year.

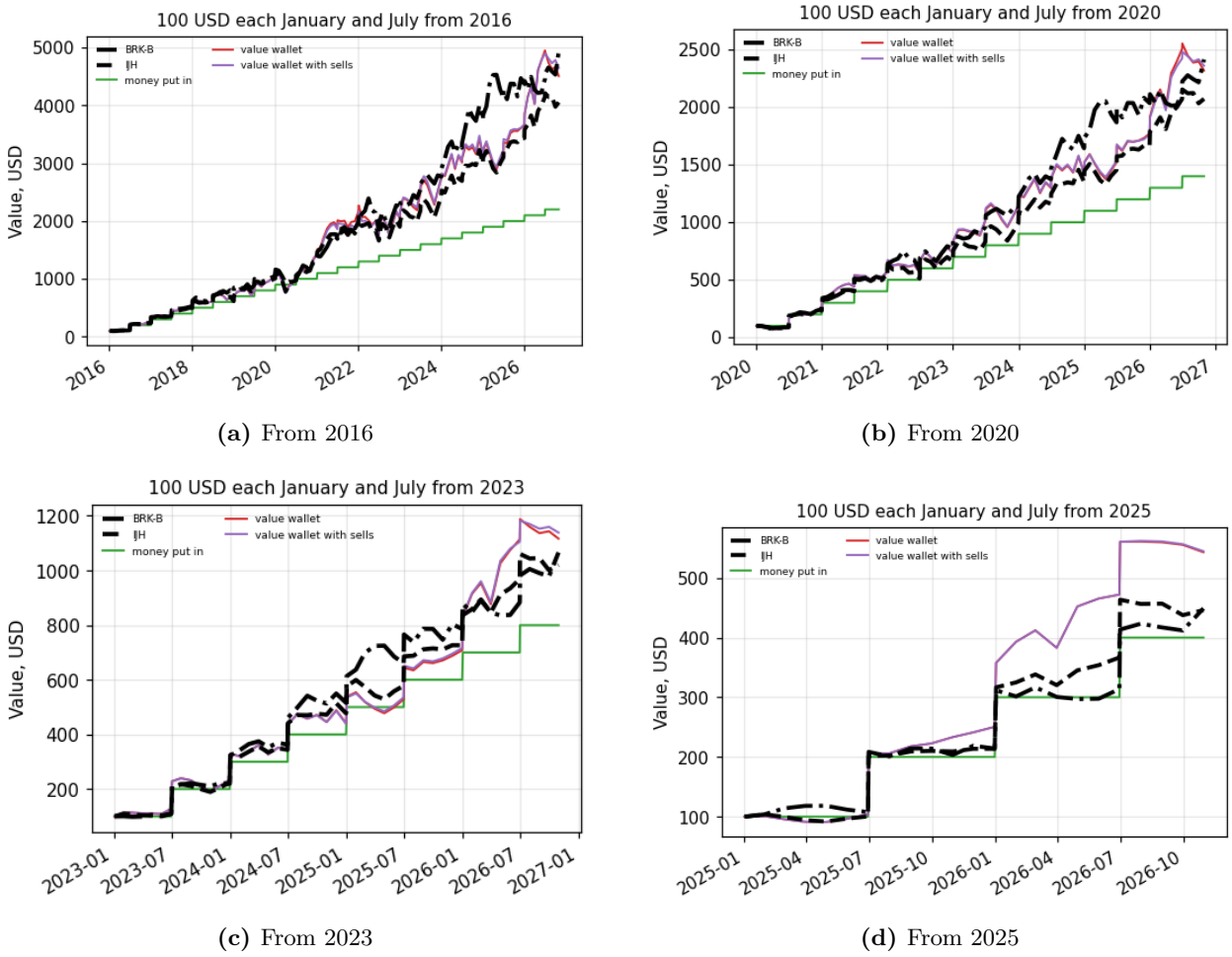
Start	Holder	Put in	Value	Multiple	A year
2016	Value wallet	2,200	4,507	2.05	12.4%
	Value wallet with sells	2,200	4,646	2.11	12.9%
	S&P MidCap 400	2,200	4,053	1.84	10.6%
	Berkshire Hathaway	2,200	4,938	2.24	13.9%
2020	Value wallet	1,400	2,318	1.66	14.0%
	Value wallet with sells	1,400	2,361	1.69	14.5%
	S&P MidCap 400	1,400	2,073	1.48	10.9%
	Berkshire Hathaway	1,400	2,415	1.73	15.2%
2023	Value wallet	800	1,116	1.40	16.5%
	Value wallet with sells	800	1,139	1.42	17.6%
	S&P MidCap 400	800	1,020	1.28	12.0%
	Berkshire Hathaway	800	1,066	1.33	14.2%
2025	Value wallet	400	543	1.36	31.4%
	Value wallet with sells	400	545	1.36	31.7%
	S&P MidCap 400	400	446	1.11	10.5%
	Berkshire Hathaway	400	449	1.12	11.1%

Table 2 and Figure 2 follow the four investors. Each puts in 100 USD every January and July from the start year; the value wallet never sells. The return a year is money-weighted: the constant yearly rate that turns the payments into today's value.

The investor who started in 2016 earns 12.4% a year against 10.6% in the S&P MidCap 400 and 13.9% in Berkshire Hathaway. From 2020 the rates are 14.0%, 10.9% and 15.2%; from 2023 16.5%, 12.0% and 14.2%; from 2025, over less than two years, 31.4%, 10.5% and 11.1%. The wallet's value swings more than either comparison, because 10 companies of a few industries are far less diversified than the index or Berkshire.

The value wallet with sells buys the same picks with the same money, but at each decision it also sells part of any holding whose score has fallen; Section 4 describes the rule. From 2016 it is worth

4,646 USD (12.9% a year), from 2020 2,361 USD (14.5%) and from 2025 545 USD (31.7%).



(a) From 2016

(b) From 2020

(c) From 2023

(d) From 2025

Figure 2: Value of 100 USD every January and July from each start year, in USD, next to the money put in; the S&P MidCap 400 dashed and Berkshire Hathaway dash-dot.

4 Selling below the threshold

Each January and July, before buying, the wallet with sells checks every company it holds. The score is put on a scale from 0 to 1, the share of the most the company could score on the rules it is measured on. A holding scoring 0.4 or less is sold in part: the share sold is (0.4 less the score) divided by 0.4, so all of it at a score of 0 and almost none just under 0.4. Holdings are sold, never sold short. The proceeds are added to the decision's 100 USD and buy its picks.

4.1 Sold in the wallet

Table 3: Sales of the investor who started in 2020: share of each holding sold, its score on the 0-to-1 scale, and the proceeds.

Year	Sold: share of the holding (score)	Proceeds, USD
2021-07	GAP 24% (score 0.30)	4.83
2022-07	GAP 44% (score 0.23), JAZZ 24% (score 0.30)	6.73
2023-01	GAP 43% (score 0.23), JAZZ 25% (score 0.30)	5.28
2023-07	GAP 41% (score 0.23), MTZ 32% (score 0.27), MMS 22% (score 0.31), JAZZ 57% (score 0.17), LEA 15% (score 0.34)	20.10
2024-01	GAP 40% (score 0.24), MTZ 3% (score 0.39), MMS 12% (score 0.35), JAZZ 57% (score 0.17), LEA 7% (score 0.37)	6.10
2024-07	MKSI 30% (score 0.28), GAP 20% (score 0.32), MTZ 54% (score 0.18), MMS 18% (score 0.33), JAZZ 64% (score 0.15), OLLI 1% (score 0.40), RH 35% (score 0.26)	38.34
2025-01	MKSI 38% (score 0.25), GAP 17% (score 0.33), MTZ 71% (score 0.12), MMS 15% (score 0.34), JAZZ 63% (score 0.15), OLLI 7% (score 0.37), RH 52% (score 0.19)	39.33
2025-07	MKSI 49% (score 0.20), MTZ 78% (score 0.09), MMS 16% (score 0.34), JAZZ 1% (score 0.40), RH 48% (score 0.21)	16.45
2026-01	MKSI 53% (score 0.19), MTZ 79% (score 0.08), MMS 5% (score 0.38), JAZZ 13% (score 0.35), LEA 3% (score 0.39), RH 38% (score 0.25), ASH 33% (score 0.27)	13.50
2026-07	MKSI 49% (score 0.20), AVNT 18% (score 0.33), MTZ 59% (score 0.16), JAZZ 43% (score 0.23), RH 67% (score 0.13), AMKR 9% (score 0.36), ASH 33% (score 0.27)	24.54

Table 3 lists the 48 sales of the investor who started in 2020, worth 175 USD in all. They are the wallet's own cash flows; the list below does not depend on what anyone holds.

4.2 Lowest scores in the index

Table 4: The 10 lowest-scoring of the companies scored on the latest score date: sector, score, score on the 0-to-1 scale and colour (Green, Amber or Red; not a recommendation).

Company	Sector	Score	Score 0 to 1	Colour
Chewy (CHWY)	Consumer Discretionary	1.2	0.09	Red
Avis Budget Group (CAR)	Industrials	1.3	0.04	Red
DigitalOcean (DOCN)	Information Technology	1.3	0.04	Red
Talen Energy (TLN)	Utilities	1.4	0.07	Red
Okta, Inc. (OKTA)	Information Technology	2.4	0.17	Red
Docusign (DOCU)	Information Technology	2.9	0.12	Red
SiriusXM (SIRI)	Communication Services	3.0	0.09	Red
UGI Corp (UGI)	Utilities	3.0	0.09	Red
Guidewire Software (GWRE)	Information Technology	3.4	0.14	Red
Hims & Hers Health (HIMS)	Health Care	3.4	0.24	Amber

These are the 10 lowest scores, as many as the picks, among the companies of the S&P MidCap 400 that are scored on 18 September 2026, whether or not anyone holds them: Chewy (CHWY),

Avis Budget Group (CAR), DigitalOcean (DOCN), Talen Energy (TLN), Okta, Inc. (OKTA), Docusign (DOCU), SiriusXM (SIRI), UGI Corp (UGI), Guidewire Software (GWRE) and Hims & Hers Health (HIMS) (Table 4). Financials, Real Estate and companies with too few rules to score are not scored and so cannot appear. Companies with the same score are ordered by symbol, so when several share the last place, the alphabetically first are shown. The list shows where the score ranks these companies today; it is not a recommendation to buy, sell or sell short anything. A score of 0 means only that no rule awarded the company any points; it says nothing about what the company is worth.

The last column colours each score on the 0-to-1 scale against the sell threshold of 0.4. Green is a score above 0.4. Amber is a score from 0.2 up to and including 0.4: a company one would not buy, and would hold only if already owned. Red is a score below 0.2: a company one might start to think about selling. The colours are not the rule of the wallet with sells. That rule is mechanical: at or below 0.4 it sells part of a holding, the more the lower the score, and at 0 all of it, whatever the colour. The sell signal on these companies has not been tested: no test has measured what the lowest scores did afterwards. The cut-offs are those the score uses today and are not fixed. The wallet never sells short; a short position on these companies is a later step and has not been tested.

5 Wallet contents

Figure 3 follows every company in the wallet with sells of the investor who started in 2020, month by month. A line starts or steps up when the company is bought, steps down when part of it is sold, and rises or falls with the price in between. The dashed lines follow the same money in the S&P MidCap 400 and in Berkshire Hathaway, divided by the number of companies the wallet holds on the date, so they show what one holding would be worth in the index or in Berkshire. Appendix B gives each holding's value after every January's trades, the money put in and the proceeds of sales.

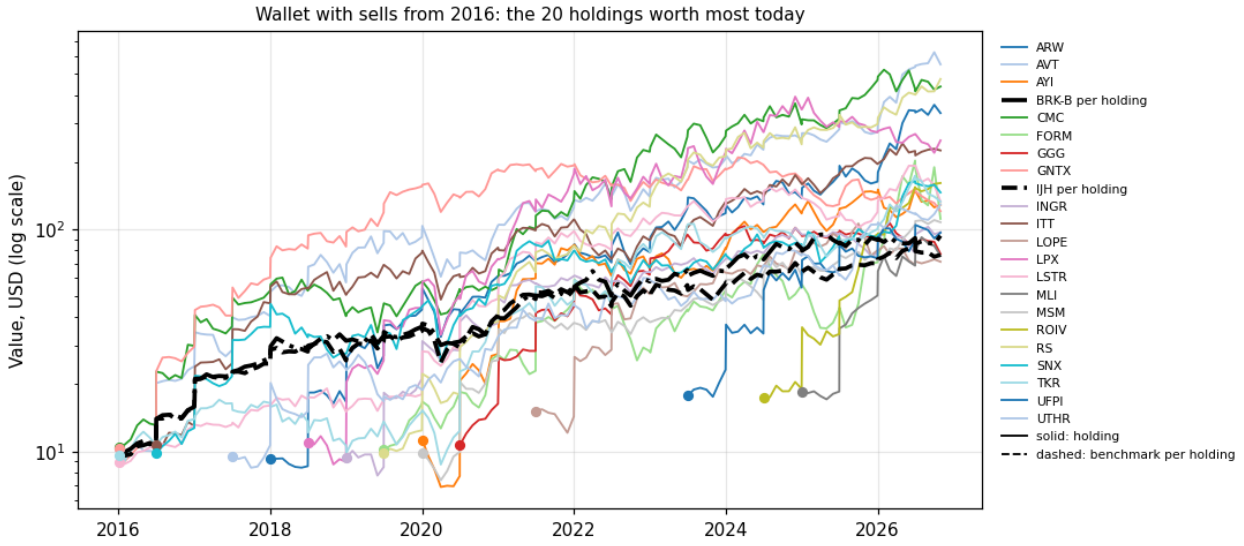


Figure 3: Value of each holding in the wallet with sells (solid) and of the S&P MidCap 400 and Berkshire Hathaway divided by the number of holdings (dashed), in USD on a logarithmic scale.

6 Rules behind the winners

Table 5: Dollars ahead of the same money in the S&P MidCap 400, per pick: total, average and median of the picks that beat the index and of those that lagged it.

Picks	Number	USD ahead, total	Average	Median
Beat the S&P MidCap 400	118	+1,070	+9.1	+5.6
Lagged the S&P MidCap 400	102	-616	-6.0	-4.7
All picks	220	+454	+2.1	+0.9

Of the picks, 118 have done better than the S&P MidCap 400 since their purchase and 102 have not, but a count treats a large win and a small loss alike. Table 5 weighs them in dollars: each pick's value today less what the same money would be worth in the S&P MidCap 400. The winners are 1,070 USD ahead of the index and the laggards 616 USD behind, a net 454 USD, the wallet's whole lead over the index. The median winner is 6 USD ahead and the median laggard 5 USD behind; the average win is 1.5 times the average loss. The wins are concentrated: the 5 largest, ITT Jul 2016, LECO Jan 2016, ITT Jan 2017, FORM Jul 2019 and RS Jul 2020, make up 16% of the winners' total.

Figure 4 compares the mean points per rule of the winners and of the laggards on the day they were picked, each pick weighted by its dollars ahead or behind the index, so the large wins and losses count most; the rules are ordered by how much more the winners scored. The winners scored most above the others on net-net (NN), P/E times P/B (G7), P/E (G6) and current ratio (G2a); the picks that lagged scored higher on return on equity (B1) and current dividend (E4). With so few picks, these differences describe this sample; they do not establish which rules work. Appendix E explains each rule, and Appendix C tests whether the rule weights favour the rules that told winners from losers.

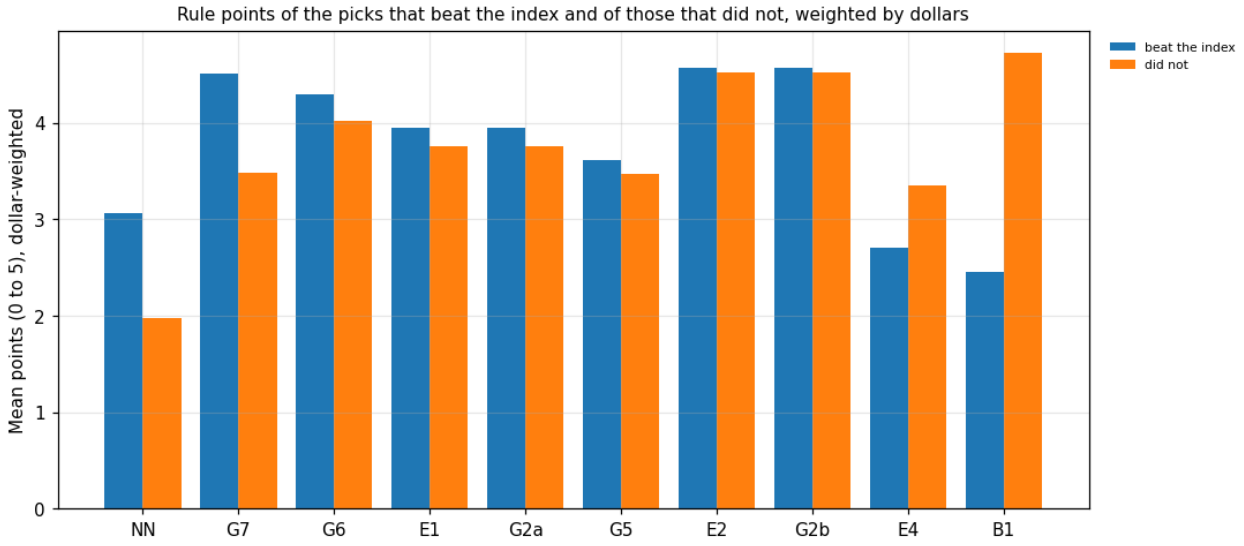


Figure 4: Mean points per rule of the picks that beat the S&P MidCap 400 since purchase and of those that did not, each pick weighted by its dollars ahead or behind the index.

7 Price history of the picks

Figure 5 shows the share price of every company the investor who started in 2023 bought or sold, and of the next picks, over the last 5 years. The price is split-adjusted and excludes dividends, as a price chart shows it; the companies run from the best to the worst price change over the 5 years,

given beside each name; when they do not fit on one page, only the 6 best and the 6 worst are shown. Each purchase is marked in green and each sale in red, at the price on the decision date; a larger marker means more money. Figure 6 gives the same trades in USD per company and decision date.

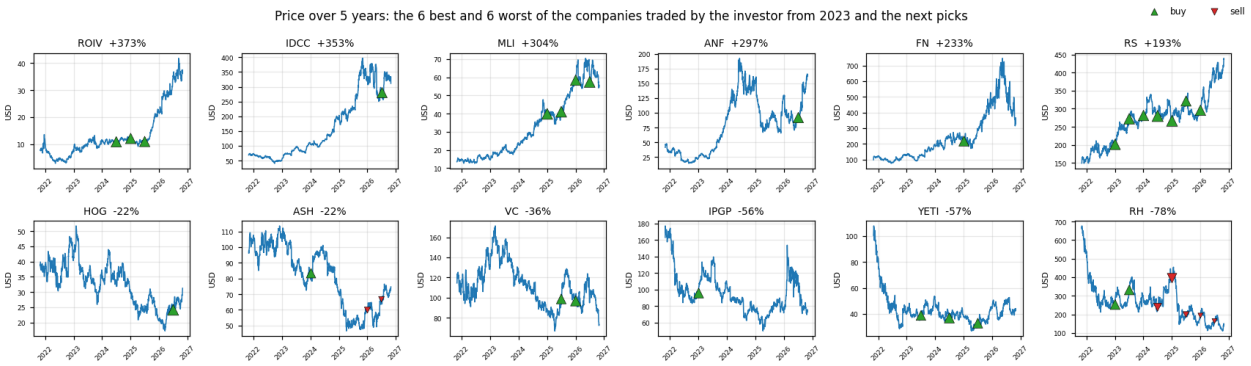


Figure 5: Share price (USD, split-adjusted, dividends excluded) over the last 5 years, one panel per company, ordered from the best to the worst price change over the period; purchases (green) and sales (red) of the investor who started in 2023, marker area in proportion to the USD traded.

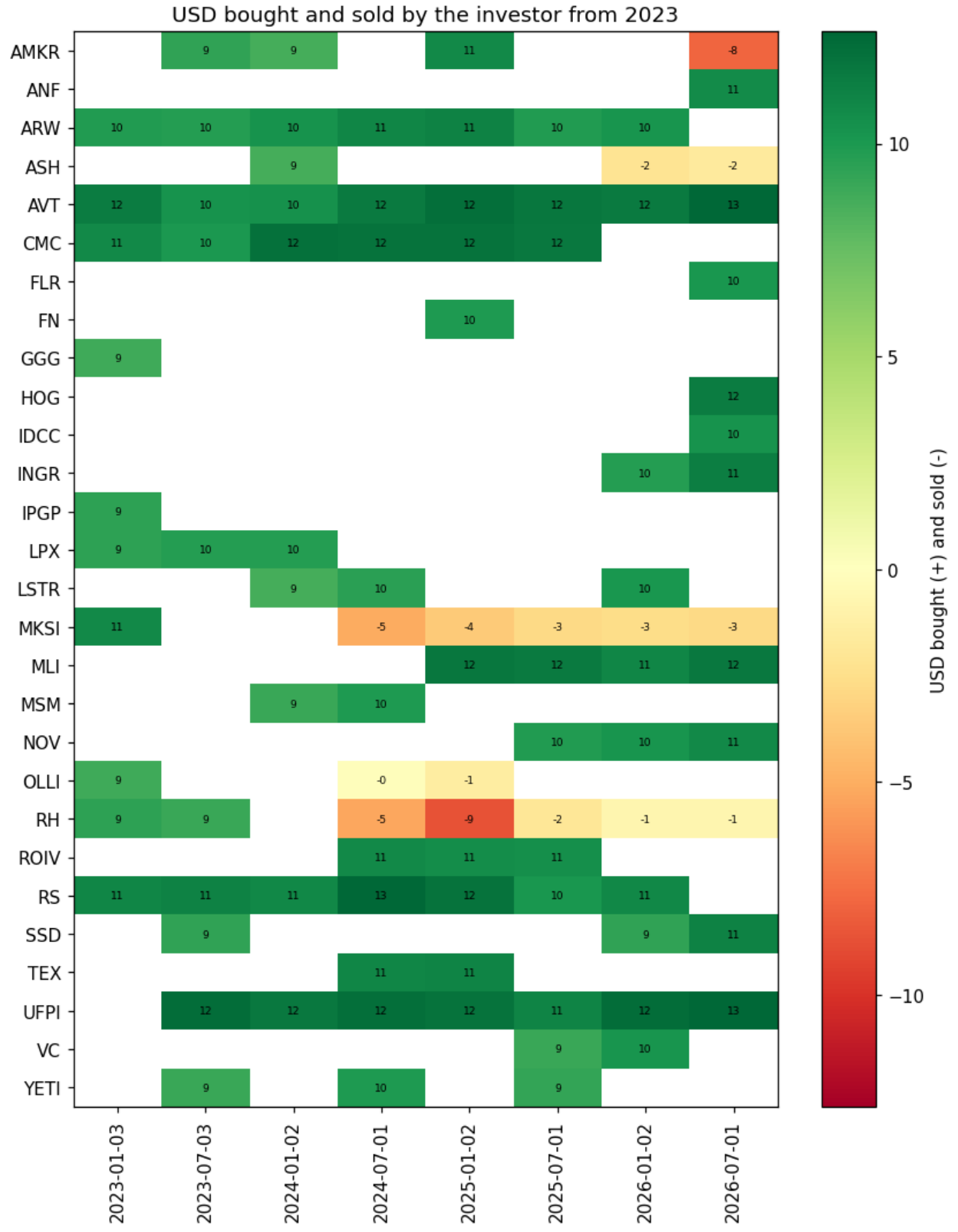


Figure 6: USD bought (green, positive) and sold (red, negative) by the investor who started in 2023, per company and decision date.

8 The next picks

Table 6: The next picks on the latest score date: score, score on the 0-to-1 scale, weight and USD of each 100 USD.

Company	Sector	Score	Score 0 to 1	Weight	USD
UFP Industries (UFPI)	Industrials	29.3	0.85	11.6%	11.57
Harley-Davidson (HOG)	Discretionary	27.0	0.78	10.6%	10.65
Ingredion (INGR)	Staples	26.7	0.77	10.5%	10.53
Simpson Manufacturing (SSD)	Industrials	26.3	0.76	10.4%	10.37
NOV Inc. (NOV)	Energy	25.3	0.73	10.0%	9.99
Fluor (FLR)	Industrials	24.6	0.78	9.7%	9.71
Commercial Metals (CMC)	Materials	24.1	0.70	9.5%	9.51
Avnet (AVT)	Technology	24.0	0.69	9.5%	9.49
Visteon (VC)	Discretionary	23.1	0.76	9.1%	9.11
InterDigital (IDCC)	Technology	23.0	0.66	9.1%	9.08

On the latest score date, 18 September 2026, the 10 highest-scoring companies are UFP Industries (UFPI), Harley-Davidson (HOG), Ingredion (INGR), Simpson Manufacturing (SSD), NOV Inc. (NOV), Fluor (FLR), Commercial Metals (CMC), Avnet (AVT), Visteon (VC) and InterDigital (IDCC) (Table 6). Each would receive its share of the next decision's 100 USD and of any proceeds from sales. The decision is taken on the first trading day of January with the scores known then, which include the annual reports filed until that day; these picks are a preview and may change. Figure 7 shows each pick's share price over the last 5 years, with the earlier purchases of the investor who started in 2023 and the potential purchase at today's price.



Figure 7: Share price of each next pick (USD, split-adjusted, dividends excluded) over the last 5 years: earlier purchases of the investor who started in 2023 in green, the potential purchase at the latest price in orange.

9 Winning picks we missed

Table 7: Winners over the next 36 months that the score missed, by index: the share of winners held (by chance), their mean score and examples with their score on a scale from 0 to 1.

Index	Winners held (by chance)	Winners' mean score	Missed winners and their score
S&P 500	10% (4%)	0.55	Amazon 0.08, Netflix 0.23, Constellation Brands 0.32, Adobe 0.48
S&P 400	5% (9%)	0.42	Paylocity 0.03, Repligen 0.16, Tenet 0.17, Qualys 0.20, Chemed 0.34
S&P 600	9% (5%)	0.51	Archrock 0.14, Brinker 0.16, SM Energy 0.19

No score picks every winner, and a winner is only known afterwards. Here the winners are the ten companies with the highest return above their index over the next 36 months, found with hindsight on every date. A book of ten companies that knew nothing would hold about 4% of them on the S&P 500, 9% on the S&P 400 and 5% on the S&P 600 (Table 7). They are shown as a check on the rules, not as advice.

On the S&P 500 the score holds 10% of the winners and ranks them slightly above the median (0.55 on a scale from 0 to 1). It found the chip makers (Nvidia 0.91, Lam Research 0.96, Micron 0.91) and missed the growth platforms: Amazon 0.08, Netflix 0.23 and Constellation Brands 0.32. On the S&P 400 the winners score below the median (0.42): Paylocity 0.03, Repligen 0.16, Tenet 0.17 and Qualys 0.20. Health care is 17% of the winners there and 3% of our picks. On the S&P 600 energy companies are 24% of the winners and 2% of our picks.

A possible cause is in the rules. They are Graham's and Buffett's, and they reward cheap, low-debt, asset-rich companies. Many winners of the period were growth and quality companies that pay no dividend and trade at higher multiples. On the S&P 400 they score 0.7 points lower than the average company on the dividend rule and 0.4 lower on debt cover, while our picks score 4.4 on debt cover against 2.1 for the average company. On the earlier sixteen-rule score, dropping the dividend rules made no measurable difference on the S&P 500 and the S&P 400, so they are not what holds the score back there; on the S&P 600 the bands excluded zero (+0.72 and -0.93). The price rules are one more possible cause: dropping them raised the best ten on the S&P 500, but that held on the S&P 500 only, showed nothing on the S&P 400 and was contradicted on the S&P 600, so it was not adopted (Section 11). Companies that S&P later added to the S&P 500 scored lower than its existing members before they joined (0.42 against 0.51 on a scale from 0 to 1), which fits the hypothesis that the score undervalues growth; it is a hypothesis, not a tested result.

10 How the model was tested

Table 8: The tests the model must pass before launch, and where each stands.

Test	What is checked	Result	Status
Winners removed	The wallet still beats its index with the 3 largest winners removed	S&P 500: yes, by 1.9 to 3.7 points (2016 to 2023 starts). S&P 400: no. S&P 600: mixed	S&P 500 passed in-sample; others not
Company samples	Repeated samples, each leaving out a random part of the companies	S&P 500: ahead of the index in every sample (the samples share most of their companies)	Passed on the S&P 500, in-sample
Unseen companies	The score frozen, then run once on a part of the companies never used	Positive, but two companies gave 99% of the gain (earlier sixteen-rule score)	To rerun
Other indices	The score beats random picks of its own index on the S&P 400 and 600	Not reproduced	Open
Two periods	The edge appears before and after 2017	Yes on the S&P 500 (earlier sixteen-rule score)	To rerun
Missed winners	Which winners the score missed, and why	Section 9	Done
Survivorship	The effect of companies that left the index	Not yet measured	Open
Real-world check	Results compared with market events and institutional logic	Not yet done	Open

This is a prototype, "first light": the first version of the model that is frozen and dated, with nothing earlier to compare it with. The figures below are in-sample stress and sampling diagnostics, compared with the index and not with random books of the same size, on a configuration that was chosen on this data after more than 100 designs were tried. The bands cover the sampling of companies only: they do not allow for the number of designs tried (multiple testing) or for the period, which the designs were chosen while looking at, so no result here is out of sample. The one held-out test, on companies the score never used, is on the earlier sixteen-rule score and is marked to rerun. The tests follow a written protocol, and a test that failed is shown as failed. The model has not cleared them: the tests below are open or failed in part. The numbers are from runs on 1 November 2026 on the ten default rules, with data as of 31 October 2026.

On the S&P 500 the picks beat the index in every one of a set of repeated samples that each left out a random part of the companies (the samples share most of their companies, so this is not independent confirmation), by 9 to 15 points a year on average depending on the start year, and by about 5 points in the worst sample. The investor who started in 2020 earns 27.7% a year against 15.9% in the index. The gain is concentrated: Micron makes up 24% and Lam Research 16% of the 2016 investor's gain. They still beat it with the three largest winners removed as if they had never existed (Applied Materials, Lam Research and Micron): by 3.7, 1.9 and 2.1 points a year for the investors who started in 2016, 2020 and 2023, and by 8.9 for 2025. With five removed (adding Nucor and Nvidia) the margin shrinks to 0.3 to 2.3 points for the first three. The edge is present in this sample of the index but thin, and concentrated in chip and materials companies that did well in a long bull market after 2009.

The other two indices do not confirm it. On the S&P 400 the picks beat their fund by 2 to 5 points a year for the 2016 to 2023 starts, and with the three largest winners removed the margin is gone (0.4, -0.4 and -0.1). On the S&P 600 the investors who started in 2016 and 2020 are ahead by about 6 points, which survives the removal of the three largest winners, while those who started in

2023 and 2025 trail the fund. The lists of the S&P 400 and 600 give no entry dates and lose the companies that grew into a larger index, so these two tests are biased and not yet a verdict on the rules. Companies that left the index are missing from every test, which flatters the results.

11 Midas and the Investor

Every result in this report was questioned before it was believed. When something looked wrong, too good, different between indices or against what a long-term investor would say, Midas, who reads the numbers, put the question to the Investor. The Investor answers only from Berkshire Hathaway's published letters to shareholders and names the year; the quotations imply no endorsement by Berkshire or Mr Buffett. Midas then tested the answer and reported back. Three of the exchanges follow. The figures in them are in-sample stress and sampling diagnostics, compared with the index and not with random books, on a configuration chosen on this data (Section 10).

Is the dividend rule holding the score back?

Midas: The winners we missed mostly pay no dividend, and they score low on the dividend rules. Should we drop them?

The Investor: (1984) "Berkshire has earned well over market rates on retained earnings, thereby creating over one dollar of market value for every dollar retained. Under such circumstances, any distribution would have been contrary to the financial interest of shareholders, large or small."

Midas: I dropped the two dividend rules from the sixteen-rule score and compared. On the S&P 500 the best ten changed by -0.03 points a year, with a 99% band from -1.05 to +0.41; on the S&P 400 by -0.26. There is no measurable difference on those two indices; on the S&P 600 the bands excluded zero (+0.72 and -0.93). The default now keeps only one dividend rule, the current dividend, and the other is a switch.

Should a cheap price be the test of value?

Midas: Dropping the P/E and price-to-book rules raised the best ten on the S&P 500 but lowered the broad ranking, and the missed winners were growth companies. That held on the S&P 500 only, and I did not adopt it.

The Investor: (1992) "Growth is always a component in the calculation of value." A low ratio of price to book value, a low price-earnings ratio or a high dividend yield are "far from determinative as to whether an investor is indeed buying something for what it is worth", and the opposite characteristics "are in no way inconsistent with a 'value' purchase". Yet "business growth, per se, tells us little about value". (2013, Graham's adage, which Buffett quotes as "Ben's adage") "Price is what you pay, value is what you get."

Midas: I tested a price-to-earnings-growth measure. On all three indices it widened the broad ranking a little and left the best ten slightly lower, within the noise. Used only for companies that pass the quality rule, return on equity, both rose on the S&P 500 and the S&P 400, by 0.3 to 1 point a year. That is small and adds complexity, so it is a switch and not the default.

Is it too good to be true?

Midas: The picks beat the index in every one of a set of repeated samples that each left out a random part of the companies. The samples share most of their companies, so that is not independent confirmation, and it still looks too good, so I removed the biggest winners as if they had never existed.

The Investor: (1987) "When shortages exist, however, even commodity businesses flourish", and "whenever shortages appear, the typical manager simply can't wait to expand capacity". (1989) "Time is the friend of the wonderful business, the enemy of the mediocre."

Midas: The 1987 passage is about insurance; I apply it to chip makers by analogy. With the three

largest winners removed (Applied Materials, Lam Research and Micron) the wallet still beat the index by 3.7, 1.9 and 2.1 points a year for the investors who started in 2016, 2020 and 2023; with five removed, by 0.3 to 2.3 points. The edge is present in this sample of the S&P 500 but thin, and it sits in chip and materials companies after 2009. On the S&P 400 it disappears without its three largest winners. Treat it with care.

What the tests showed. The results are for holding over years, not months: the picks are bought twice a year and not sold. The concentration is the main risk, and one or two big winners carry much of the gain, so they are treated as luck until tested the same way. The book holds ten companies, not five: ten was chosen for plausibility and diversification, although the five-company book scored higher in the tests. The tests cover the default ten rules; other switches were not tested as a set. This is what the tests showed, not advice; read the tests in Section 10 before relying on a number.

12 Where to read more

Table 9: Strategy literature by risk profile.

Profile	Idea	Reading
Defensive	Preserve capital; diversified, cheap, boring; index funds	Graham, The Intelligent Investor (the defensive investor); Bogle, The Little Book of Common Sense Investing
Core	A wonderful company at a fair price, held for years	The Berkshire Hathaway shareholder letters; Dorsey, The Little Book That Builds Wealth
Value with a margin of safety	Buy at a deep discount to value	Graham (the enterprising investor); Klarman, Margin of Safety
Quality growth	Growth at a reasonable price, good management	Fisher, Common Stocks and Uncommon Profits; Lynch, One Up on Wall Street
Systematic value	Rules, ranked	Greenblatt, The Little Book That Beats the Market; Piotroski (2000), the F-score; Novy-Marx (2013), gross profitability
Risk awareness	Cycles and what can go wrong	Marks, The Most Important Thing; Lakonishok, Shleifer and Vishny (1994), contrarian investment

Planned, and not yet available: a member will be able to bring the strategy literature they like and ask for answers drawn from those pages, with the source named. The Berkshire Hathaway letters are free to read on the company's site. Books are copyrighted: use your own copy. The list below is grouped by the kind of risk each approach accepts. It is a starting point only: the titles are given from general knowledge and have not been checked against the books, so check each one before relying on it.

13 Conclusions

In this sample, the value picks bought twice a year and held are worth more today than the same money in the S&P MidCap 400 or in Berkshire Hathaway for each of the four start years. That describes the past under rules and settings chosen while looking at it; it is not a test of the strategy. The lead is uneven: the 12 purchases of Reliance, Inc. (RS) make up 14% of the 2020 investor's value, and the purchases of January 2017, January 2019, July 2023 and January 2024 trail the index.

Before the strategy is used, three steps are needed: the companies that left the index must be added to the data, the score must be tested on more decisions (each quarter instead of each year), and trading costs and taxes must be included.

A Every purchase

Table 10: Every purchase of the wallet with sells of the investor who started in 2020: weight in the decision's purchase, USD invested (the decision's 100 USD and the proceeds of sales), split-adjusted price at purchase and today, USD sold from it later, USD it keeps today and its total return with dividends, (sold + kept) / invested - 1.

Year	Company	Sector	Weight	In	Bought at	Now	Sold	Kept	Return
2020-01	LPX	Industrials	12%	12	29.88	73.31		33	+171%
2020-01	MKSI	Technology	11%	11	111.83	202.04	14	1	+31%
2020-01	AYI	Industrials	10%	10	139.61	318.14		24	+133%
2020-01	GAP	Discretionary	10%	10	17.28	22.32	10	1	+6%
2020-01	AVT	Technology	10%	10	42.72	88.20		24	+144%
2020-01	ARW	Technology	9%	9	85.79	208.34		23	+143%
2020-01	LSTR	Industrials	9%	9	113.44	142.70		14	+45%
2020-01	INGR	Staples	9%	9	92.95	96.97		12	+27%
2020-01	RS	Materials	9%	9	119.25	436.91		38	+312%
2020-01	MSM	Industrials	9%	9	78.77	121.02		20	+122%
2020-07	AYI	Industrials	12%	12	92.06	318.14		41	+252%
2020-07	AVT	Technology	11%	11	26.90	88.20		42	+282%
2020-07	RS	Materials	10%	10	92.97	436.91		54	+421%
2020-07	LSTR	Industrials	10%	10	111.23	142.70		15	+45%
2020-07	AVNT	Materials	10%	10	25.46	42.84	3	16	+91%
2020-07	TKR	Industrials	10%	10	44.31	110.90		26	+178%
2020-07	MSM	Industrials	9%	9	71.63	121.02		21	+123%
2020-07	GGG	Industrials	9%	9	47.42	67.44		14	+54%
2020-07	MTZ	Industrials	9%	9	43.24	186.06	26	0	+183%
2020-07	INGR	Staples	9%	9	81.14	96.97		13	+43%
2021-01	LSTR	Industrials	11%	11	133.09	142.70		13	+21%
2021-01	RS	Materials	11%	11	119.71	436.91		43	+300%
2021-01	AYI	Industrials	10%	10	120.22	318.14		27	+169%
2021-01	CMC	Materials	10%	10	20.45	63.98		34	+240%
2021-01	MMS	Industrials	10%	10	73.07	63.72	7	4	+9%
2021-01	JAZZ	Health care	10%	10	160.40	215.98	9	0	-13%
2021-01	INGR	Staples	10%	10	78.86	96.97		14	+45%
2021-01	MSM	Industrials	10%	10	81.68	121.02		18	+84%
2021-01	LEA	Discretionary	9%	9	157.44	118.44	2	6	-12%
2021-01	GGG	Industrials	9%	9	70.60	67.44		10	+3%
2021-07	LPX	Industrials	11%	12	60.50	73.31		15	+30%
2021-07	LOPE	Discretionary	11%	11	90.88	147.88		19	+63%
2021-07	FORM	Technology	11%	11	36.75	87.53		26	+138%
2021-07	RS	Materials	10%	10	152.92	436.91		32	+211%
2021-07	LSTR	Industrials	10%	10	160.11	142.70		10	-1%
2021-07	AYI	Industrials	10%	10	176.26	318.14		19	+83%
2021-07	GGG	Industrials	10%	10	75.89	67.44		10	-5%
2021-07	CMC	Materials	10%	10	31.14	63.98		22	+121%
2021-07	MMS	Industrials	9%	10	88.11	63.72	6	3	-10%
2021-07	TKR	Industrials	9%	10	80.30	110.90		15	+51%
2022-01	CMC	Materials	12%	12	36.34	63.98		22	+88%
2022-01	OLLI	Discretionary	11%	11	52.41	90.26	2	17	+75%
2022-01	LOPE	Discretionary	11%	11	87.47	147.88		18	+69%
2022-01	FORM	Technology	10%	10	46.74	87.53		18	+87%
2022-01	TKR	Industrials	10%	10	69.15	110.90		17	+74%
2022-01	MKSI	Technology	10%	10	177.49	202.04	7	1	-19%
2022-01	LPX	Industrials	10%	10	77.99	73.31		10	+0%
2022-01	AVT	Technology	9%	9	41.10	88.20		23	+141%
2022-01	GGG	Industrials	9%	9	78.68	67.44		9	-9%
2022-01	JAZZ	Health care	9%	9	131.01	215.98	10	0	+6%
2022-07	LOPE	Discretionary	12%	12	95.37	147.88		19	+55%

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Year	Company	Sector	Weight	In	Bought at	Now	Sold	Kept	Return
2022-07	RS	Materials	11%	12	171.07	436.91		32	+173%
2022-07	CMC	Materials	11%	12	33.81	63.98		23	+100%
2022-07	MKSI	Technology	10%	11	96.32	202.04	15	1	+49%
2022-07	RH	Discretionary	10%	11	220.70	146.66	13	0	+24%
2022-07	LPX	Industrials	10%	10	54.58	73.31		15	+43%
2022-07	TKR	Industrials	9%	10	53.50	110.90		22	+122%
2022-07	ARW	Technology	9%	10	111.76	208.34		18	+86%
2022-07	GGG	Industrials	9%	10	59.50	67.44		12	+20%
2022-07	LSTR	Industrials	9%	9	146.91	142.70		10	+6%
2023-01	AVT	Technology	12%	12	41.59	88.20		28	+132%
2023-01	RS	Materials	11%	12	201.39	436.91		27	+130%
2023-01	CMC	Materials	11%	11	47.83	63.98		16	+41%
2023-01	MKSI	Technology	11%	11	84.67	202.04	18	1	+69%
2023-01	ARW	Technology	10%	10	106.14	208.34		20	+96%
2023-01	RH	Discretionary	9%	10	253.28	146.66	11	0	+8%
2023-01	IPGP	Technology	9%	10	96.49	72.59		7	-25%
2023-01	LPX	Industrials	9%	10	59.37	73.31		13	+30%
2023-01	GGG	Industrials	9%	9	67.90	67.44		10	+5%
2023-01	OLLI	Discretionary	9%	9	46.77	90.26	2	17	+96%
2023-07	UFPI	Industrials	12%	15	96.50	81.92		13	-11%
2023-07	RS	Materials	11%	13	272.16	436.91		23	+69%
2023-07	AVT	Technology	10%	12	50.09	88.20		24	+90%
2023-07	CMC	Materials	10%	12	53.47	63.98		15	+25%
2023-07	ARW	Technology	10%	12	143.95	208.34		17	+45%
2023-07	LPX	Industrials	10%	12	74.59	73.31		12	+3%
2023-07	AMKR	Technology	9%	11	29.54	39.50	3	14	+54%
2023-07	SSD	Industrials	9%	11	136.35	193.34		16	+45%
2023-07	YETI	Discretionary	9%	11	38.85	43.66		12	+12%
2023-07	RH	Discretionary	9%	11	331.79	146.66	9	0	-17%
2024-01	CMC	Materials	12%	13	50.18	63.98		17	+33%
2024-01	UFPI	Industrials	12%	12	123.56	81.92		9	-31%
2024-01	RS	Materials	11%	12	281.43	436.91		19	+62%
2024-01	AVT	Technology	10%	11	49.18	88.20		21	+91%
2024-01	ARW	Technology	10%	11	119.89	208.34		19	+74%
2024-01	LPX	Industrials	10%	10	69.73	73.31		11	+9%
2024-01	MSM	Industrials	9%	10	99.50	121.02		13	+35%
2024-01	LSTR	Industrials	9%	9	189.43	142.70		7	-21%
2024-01	AMKR	Technology	9%	9	32.00	39.50	2	11	+42%
2024-01	ASH	Materials	9%	9	83.70	74.68	4	4	-14%
2024-07	RS	Materials	11%	16	279.98	436.91		26	+62%
2024-07	UFPI	Industrials	11%	15	110.32	81.92		12	-23%
2024-07	CMC	Materials	11%	15	53.52	63.98		18	+23%
2024-07	AVT	Technology	11%	15	50.72	88.20		27	+83%
2024-07	ARW	Technology	10%	14	119.27	208.34		24	+75%
2024-07	TEX	Industrials	10%	14	53.58	56.15		15	+8%
2024-07	ROIV	Health care	10%	13	10.83	36.91		46	+241%
2024-07	MSM	Industrials	9%	12	78.23	121.02		21	+69%
2024-07	YETI	Discretionary	9%	12	36.94	43.66		15	+18%
2024-07	LSTR	Industrials	9%	12	183.40	142.70		10	-18%
2025-01	AVT	Technology	11%	15	51.67	88.20		27	+78%
2025-01	CMC	Materials	11%	15	49.15	63.98		20	+34%
2025-01	RS	Materials	11%	15	266.61	436.91		25	+68%
2025-01	UFPI	Industrials	11%	15	111.16	81.92		11	-24%
2025-01	MLI	Industrials	10%	14	39.87	54.56		20	+40%

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Year	Company	Sector	Weight	In	Bought at	Now	Sold	Kept	Return
2025-01	ARW	Technology	10%	14	111.38	208.34		26	+87%
2025-01	TEX	Industrials	10%	14	44.52	56.15		18	+29%
2025-01	AMKR	Technology	9%	13	26.01	39.50	4	19	+70%
2025-01	ROIV	Health care	9%	13	12.08	36.91		40	+206%
2025-01	FN	Technology	9%	12	220.37	327.29		18	+49%
2025-07	AVT	Technology	11%	13	53.93	88.20		22	+68%
2025-07	CMC	Materials	11%	13	50.71	63.98		17	+28%
2025-07	MLI	Industrials	11%	13	41.03	54.56		17	+35%
2025-07	UFPI	Industrials	11%	12	105.47	81.92		10	-21%
2025-07	ROIV	Health care	10%	12	10.94	36.91		40	+237%
2025-07	RS	Materials	10%	11	321.33	436.91		16	+39%
2025-07	ARW	Technology	9%	11	129.76	208.34		17	+61%
2025-07	NOV	Energy	9%	11	12.92	19.29		17	+54%
2025-07	YETI	Discretionary	9%	10	32.93	43.66		14	+33%
2025-07	VC	Discretionary	9%	10	98.77	73.32		8	-25%
2026-01	UFPI	Industrials	12%	13	92.18	81.92		12	-10%
2026-01	AVT	Technology	11%	12	49.34	88.20		23	+81%
2026-01	MLI	Industrials	10%	12	58.40	54.56		11	-6%
2026-01	RS	Materials	10%	12	295.82	436.91		17	+49%
2026-01	ARW	Technology	10%	11	113.04	208.34		20	+84%
2026-01	VC	Discretionary	10%	11	96.88	73.32		8	-23%
2026-01	NOV	Energy	10%	11	16.41	19.29		13	+20%
2026-01	LSTR	Industrials	10%	11	146.78	142.70		11	-1%
2026-01	INGR	Staples	9%	10	109.79	96.97		9	-9%
2026-01	SSD	Industrials	9%	10	164.72	193.34		12	+18%
2026-07	UFPI	Industrials	11%	14	89.15	81.92		13	-8%
2026-07	AVT	Technology	11%	14	87.28	88.20		14	+1%
2026-07	MLI	Industrials	10%	13	57.42	54.56		12	-5%
2026-07	HOG	Discretionary	10%	13	24.15	31.22		17	+30%
2026-07	INGR	Staples	10%	13	95.62	96.97		13	+2%
2026-07	SSD	Industrials	10%	12	203.42	193.34		12	-5%
2026-07	NOV	Energy	10%	12	18.10	19.29		13	+7%
2026-07	ANF	Discretionary	9%	12	92.32	164.44		21	+78%
2026-07	IDCC	Technology	9%	11	282.26	311.57		13	+11%
2026-07	FLR	Industrials	9%	11	50.35	46.86		10	-7%

B Wallet contents by year

Table 11: Wallet with sells of the investor who started in 2020: value of each holding after each January's trades and today (USD), and the money put in and the proceeds of sales to date, July decisions included.

Holding	2020	2021	2022	2023	2024	2025	2026	Now
MKSI	11	15	28	35	41	19	7	4
ARW	9	11	15	31	56	79	100	185
AVT	10	23	37	50	84	121	143	273
AYI	10	35	74	58	71	104	131	112
LSTR	9	35	59	66	87	90	90	90
LPX	12	15	58	66	100	150	120	109
RS	9	34	57	98	165	188	235	350
GAP	10	11	8	2	1	1	1	1
MSM	9	32	34	34	53	54	64	93
INGR	9	27	34	36	41	53	54	62

Table 11: Wallet with sells of the investor who started in 2020: value of each holding after each January's trades and today (USD), and the money put in and the proceeds of sales to date, July decisions included.

Holding	2020	2021	2022	2023	2024	2025	2026	Now
GGG		23	46	61	78	77	77	64
TKR		16	33	47	55	49	62	80
AVNT		15	22	14	17	17	14	16
MTZ		14	20	19	11	3	0	0
JAZZ		10	17	12	2	0	0	0
LEA		9	11	8	7	5	6	6
CMC		10	42	84	113	141	227	205
MMS		10	20	19	15	10	9	7
FORM			24	11	21	23	30	45
LOPE			22	40	50	61	62	56
OLLI			11	19	30	40	42	34
RH				22	34	15	2	1
IPGP				10	11	7	8	7
SSD					16	14	24	40
UFPI					31	59	73	79
AMKR					21	31	52	43
YETI					14	23	42	40
ASH					9	8	5	4
ROIV						28	74	125
TEX						25	32	32
FN						12	26	18
MLI						14	52	61
VC							21	16
NOV							25	43
IDCC								13
HOG								17
FLR								10
ANF								21
Put in	100	300	500	700	900	1,100	1,300	1,400
Sold	0	0	5	17	43	121	151	175
Total	100	346	672	841	1,234	1,521	1,910	2,361

C Testing the weighting

Table 12: Each rule's weight w_r on the picks' dates (overall, for the picks that beat the S&P MidCap 400 and for the others), the dollar-weighted mean points, the gap and the rank correlation with the return a year above the S&P MidCap 400.

Rule	w_r	w_r beat	w_r not	Pts beat	Pts not	Gap	Corr.
NN	1.00	1.00	1.00	3.06	1.97	+1.09	+0.29
B1	0.85	0.85	0.85	2.45	4.72	-2.27	-0.51
G7	0.84	0.84	0.84	4.51	3.49	+1.03	+0.39
G6	0.82	0.82	0.82	4.30	4.02	+0.28	+0.20
G2b	0.69	0.69	0.69	4.57	4.52	+0.05	+0.00
E2	0.66	0.66	0.66	4.57	4.52	+0.05	+0.00
G2a	0.61	0.61	0.61	3.95	3.76	+0.19	+0.05
G5	0.55	0.55	0.56	3.61	3.47	+0.14	-0.01
E1	0.37	0.38	0.37	3.95	3.76	+0.19	+0.05
E4	0.32	0.33	0.32	2.70	3.35	-0.65	-0.03

Each rule's points are multiplied by its weight, one less the share of companies passing it on the date, so a rule few companies pass counts more. A good weighting gives most weight to the rules whose points went with better returns. Table 12 sets each rule's mean weight on the dates the picks

were made against two measures of how well it told the 220 picks apart: the gap between the mean points of the picks that beat the S&P MidCap 400 and of those that did not, weighted by dollars as in Figure 4, and the rank correlation of the rule's points with each pick's return a year above the S&P MidCap 400. Figure 8 plots weight against gap; a good weighting puts its rules on a rising line, with the heavy rules above zero.

Across the rules, the rank correlation of weight and gap is +0.35, and of weight and correlation +0.39.

The weighting leans towards the rules that worked, though far from fully. Weighted heavily but with a negative gap: return on equity (B1). A positive gap but a light weight: current ratio (G2a), 6-year earnings growth (G5), current ratio (lenient) (E1) and debt to working capital (lenient) (E2). A rule's weight hardly differs between the dates of winning and losing picks (columns w_r beat and w_r not), because it is set across all companies on a date, not per pick. These differences suggest where the weighting could be tuned; any change must be tested on years it was not fitted on.

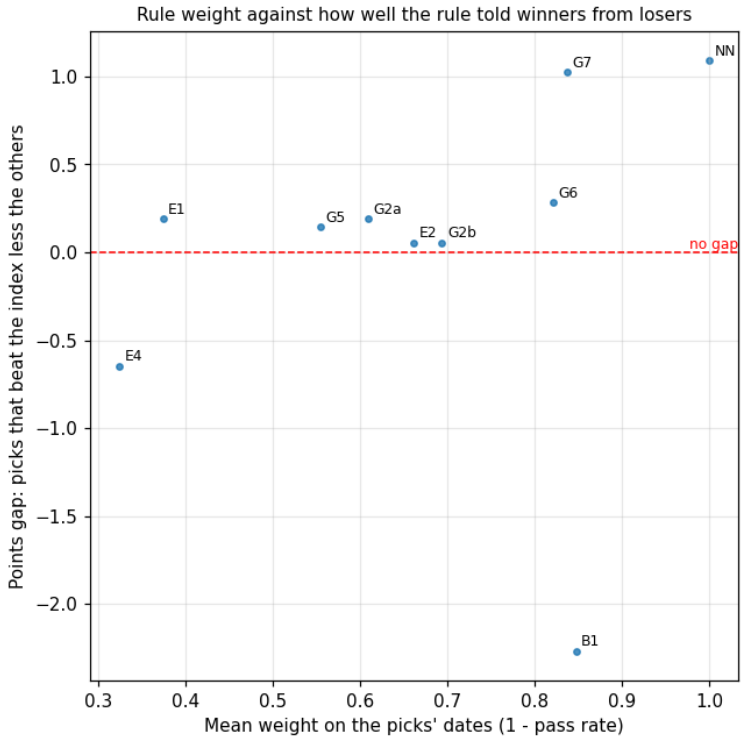


Figure 8: Each rule's mean weight against the gap in points between the picks that beat the S&P MidCap 400 and the others, weighted by dollars; rules above the line told winners from losers.

D Forward information

Table 13: Truncation check: companies scored on the date, companies whose score differs when every input is cut at the date (per score), and whether the picks are the same.

Date	Companies	Score	Score 0 to 1	Unweighted	Share passed	Same picks
2018-12-21	128	0	0	0	0	yes
2022-12-16	187	0	0	0	0	yes
2025-12-19	272	0	0	0	0	yes

A score must use only what was known on its date. Four guards are built in. A company's figures count from the day its annual report was filed with the SEC, and of all the figures filed for a year only the first reported is used, never a later restatement. Prices are the close on the scoring date

and dividends are counted by their ex-date. Each January and July purchase uses the scores of the rebalance date two weeks before it. Rule weights come from the companies on the date alone.

These guards are tested by cutting every input at a date as if it were that day (filings filed on or before it, prices and dividends up to it), scoring the companies again and comparing the scores with the full run (Table 13). Any difference means information from after the date reached a score. On 21 December 2018, 16 December 2022 and 19 December 2025 the picks were the same, and the scores were identical on 21 December 2018, 16 December 2022 and 19 December 2025. The same check runs as a unit test. A rule whose window reaches back further than a company's price history (a long dividend record of a younger company) is not evaluated rather than failed.

Three sources of forward information are outside this test. Only today's S&P MidCap 400 members are scored. The index's list gives no entry dates: a company counts from its latest addition in the list of changes, or from the start of that list (January 2012) when no addition is listed. The list holds selected changes only, so some companies count before they joined: a look-ahead that flatters the results. Companies that later left the index are missing, which flatters every result. Sectors are today's, used to leave out Financials and Real Estate in earlier years. And the score's design and settings were chosen while looking at 2009 to 2026, so no period tests it fairly: 2022 onwards, once meant as a held-out test, was used to choose between models.

D.1 Companies missing from the data

Table 14: Companies that left the S&P MidCap 400 since each start year, all missing from the data: in all, on their market value, and through an acquisition or another reason.

Start	Left the index	On market value	Acquired or other
2016	468	217	251
2020	293	203	90
2023	164	123	41
2025	65	41	24

Table 14 counts the companies that left the S&P MidCap 400 after each start year; none of them is in the data, of any sector. Most left through an acquisition, often at a premium; those that dropped out on their market value are the likely losers a real investor could have picked. The later the start, the fewer are missing, which is why the investor who started in 2020 is the reference of this report. The count starts with the index's list of changes, which begins in January 2012: removals before that are not counted, so for earlier start years it understates the gap. The bias is removed only when the historical members of the index are added to the data.

E The rules

Table 15: The rules: what each measures, the better direction and when a company passes.

Rule	Name	What it measures	Better	Passes when
G2a	current ratio	Current assets divided by current liabilities: what the company owns against what it owes within a year.	higher	at least 2.0
G2b	debt to working capital	Long-term debt divided by net current assets (current assets less current liabilities).	lower	at most 1.0
G5	6-year earnings growth	Average earnings per share of the last 3 years over the average of the 3 years before them, within a window of 6 years.	higher	at least 1.33 times the earlier average
G6	P/E	Price over the average earnings per share of the last 3 years.	lower	at most 15.0
G7	P/E times P/B	P/E as in G6 times the price over book value per share.	lower	P/B at most 1.5, or the product at most 22.5
E1	current ratio (lenient)	Current ratio, as G2a.	higher	at least 1.5
E2	debt to working capital (lenient)	Long-term debt over net current assets, as G2b.	lower	at most 1.1
E4	current dividend	Dividends paid over the last 365 days, as a yield on the price.	higher	any dividend
NN	net-net	Market value over net current asset value (current assets less all liabilities).	lower	at most 0.67
B1	return on equity	Number of the last 3 years with return on equity (net income over equity) of at least 15%.	higher	at least 2 of those years, and long-term debt at most 0.5 times equity

Table 15 explains each rule: what it measures, whether a higher or a lower value earns more points, and when a company passes it. The points come from the company's rank on the measure among all companies on the date; the pass threshold sets the rule's weight. Rules G follow Graham's defensive investor (The Intelligent Investor, chapter 14), rules E his enterprising investor (chapter 15) and his net-net test (NN), B1 Buffett's return on equity, and NI1 was added for net income growth. Two pairs test the same measure at different strictness (G2a and E1, G2b and E2), so the current ratio and debt count twice.